

Software Engineering - Project Proposal

Collaborative Study Material & Quiz Platform

Group 5

Course: Software Engineering (UE24CS341A)

Faculty: Dr. Sudeepa Roy Dey, Associate Professor, Dept. of CSE, PES University

Team Profile

Name	SRN
Bharat Shandilya	PES2UG24CS906
Sandeep H	PES2UG24CS440
Samyak Sanklecha	PES2UG24CS436
Shreyas B Patil	PES2UG24CS488

Every team member is involved in all project activities. The ownership split below is only for the functional features each of us leads.

Project Description

For our mini-project we are building a Collaborative Study Material & Quiz Platform - a web application where students can share study material, build a common question bank, create and attempt quizzes, and track how they are doing. We picked this idea because as students we often keep our notes and practice questions scattered across different apps, and there is no single place where a group can prepare together and also test themselves. Our typical users are students of a class or a study group who want to prepare collaboratively and check their progress.

We are building it on the MERN stack (MongoDB, Express.js, React, Node.js). For this version we plan to run and demonstrate it on localhost rather than deploying it publicly. We do not have an external customer; it is meant for student use.

By the end of the project, a user will be able to:

- Upload, share and search study materials organised by subject and topic.
- Contribute questions to a shared, topic-based question bank.

- Create quizzes from the question bank and attempt them, with objective questions graded automatically.
- View scores and topic-wise performance to track improvement over time.
- Give feedback and ratings on materials and questions to highlight the useful ones.

Plan of Work and Product Ownership

We have split the five functional features across the team, with Samyak taking two of the lighter ones. The Administration part (managing users, content and topics) is a shared platform-level task we maintain together, along with the common work - UI, database design, testing and integration.

Member	Functional Feature Owned
Samyak Sanklecha	Collaborative Study Material Sharing and Collaborative Content Evaluation and Feedback - upload/browse of materials and the ratings/feedback that surface good resources.
Sandeep H	Topic-Based Question Bank - the shared question bank (add/organise questions by subject, topic, difficulty and type, with search/filter) that quizzes draw from.
Bharat Shandilya	Quiz Creation and Attempting - building quizzes from the question bank, attempting them and auto-evaluating objective questions.
Shreyas B Patil	Performance Analysis and Progress Tracking - scores, accuracy, topic-wise performance and improvement over time.

Short-term plan (next few weeks):

- Finalise the database schema and set up the MERN project skeleton together.
- Each member implements their feature module and its screens.
- Integrate the modules, test the main flows and fix issues before the demo.

We will use GitHub for version control and Jira to plan and track our tasks and sprints through the project.